Ringel's Conjecture

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Chapter 1

Formal Status

The public theorem in `Ringel/Statement.lean` has the form

`CaseABSourceStatement ringel_conjecture_large`.

The source package records the Case A and Case B statements needed by the proof. The deterministic assembly and the Case C branch are formalized in Lean. The unconditional probability, repletion, exact-colour matching, and path-absorption arguments for Cases A and B remain the principal source-level boundary.

Chapter 2

Statement

Definition 2.1. Ringel's conjecture for all n. This remains open.

Definition 2.2. The large-n theorem, conditional on .

Chapter 3

Combinatorial Primitives (Primitives.lean)

Definition 3.1. Near-difference rainbow colouring of K_{2n+1}.

Theorem 3.2. *ND-colouring is a 2-factorization.*

Lemma 3.3. *A step by* $\acute{s}(c+1)$ *realizes colour c.*

Definition 3.4. The three structural case predicates.

Chapter 4

Tree Structure and Case Division

Theorem 4.1. *A tree splits into many leaves or many vertex-disjoint bare paths.*

Theorem 4.2. *Connects the split construction to the Case A/B/C division.*

Theorem 4.3. *Routes an admissible tree into Case A, Case B, or Case C.*

Chapter 5

Case A Source Interface (CaseA.lean and CaseASource.lean)

Definition 5.1. Signed walk sum in the cyclic vertex set.

Lemma 5.2. *Embeds a tree through signed edge increments.*

Definition 5.3. Records the core embedding, leaf positions, injectivity, and rainbow conditions needed by the Case A finishing step.

Definition 5.4. Records a finite Case A output together with its core map, reservoirs, and perfect rainbow matching.

Theorem 5.5. *The eventual Case A source statement used by the public source package.*

Chapter 6

Case B Source Interface (CaseB.lean and CaseSource.lean)

Definition 6.1. The eventual Case B source statement used by the public source package.

The fixed-endpoint path construction, its random availability estimates, and exact collective colour use remain to be formalized from the MPS source.

Chapter 7

Case C — Small Core (CaseC.lean)

Lemma 7.1. *Constructs the Case C rainbow embedding for sufficiently large n.*

Theorem 7.2. *Produces a rainbow copy in Case C.*

Chapter 8

Probability and Reservoir Infrastructure

Definition 8.1. Finite probability space used by the formal development.

Lemma 8.2. *Positive probability implies existence.*

The near-embedding and reservoir modules formalize finite sample spaces, concentration bookkeeping, and retained sets. They do not by themselves prove the full MPS Case A/B source statements.

Chapter 9

Proof Assembly (Spine.lean)

Theorem 9.1. *The Kotzig cyclic-shift construction turns one rainbow copy into an edge decomposition.*

Theorem 9.2. *The source package yields an eventual rainbow copy.*

Theorem 9.3. *Assembles the large-n theorem from the source package.*